

1. Student Details

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About Me: I am a student pursuing the BS Degree program at IIT Madras with an interest in web application development and database management. I enjoy developing practical web-based applications that solve real-world problems. This project helped me gain hands-on experience with Flask, SQLAlchemy, and database-driven web development.

2. Project Details

Project Title

Adventure Trekking Management System

Problem Statement

To design and develop a web-based Adventure Trekking Management System that enables administrators to manage trekking routes, staff members, users, and bookings efficiently. The system also allows users to browse available treks, book trekking adventures, and monitor their booking history, while staff members can manage their assigned trekking activities and profiles.

Approach

The application is developed using the Flask web framework following a modular architecture. It provides separate dashboards for administrators, staff members, and users. The system allows administrators to create and manage trekking routes, assign staff, manage users, and monitor bookings. Users can register, log in, explore available treks, make bookings, and view their trekking history. Staff members can manage their profiles and monitor assigned treks.

SQLite is used as the backend database with SQLAlchemy ORM to manage database operations and relationships efficiently.

3. AI / LLM Declaration

I used ChatGPT (OpenAI) to assist in understanding Flask concepts, improving HTML and CSS layouts, resolving SQLAlchemy relationship issues, fixing template errors, and enhancing documentation. The extent of AI/LLM usage is approximately **10–15%**, primarily for guidance, debugging support, UI improvements, and documentation.

The complete project design, implementation, testing, integration, and final code were developed and verified manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Backend web framework
Python	Core programming language
SQLAlchemy	ORM for database operations
SQLite	Database management
HTML5	Structure of web pages
CSS3	Styling and user interface
Jinja2	Dynamic HTML template rendering
Werkzeug Security	Password hashing and authentication

5. Database Schema / ER Diagram

Tables

AdventureUser

Stores user and staff account information.

Fields:

- `usr_ref_id`

- username
 - usr_mail
 - phone_no
 - county
 - city_of_usr
 - password
 - rolee
 - is_it_active
 - staf_satus
-

Adventure_Staff

Stores additional information related to staff members.

Fields:

- staf_ref_id
 - usr_ref_id
 - staf_nam
 - skil_area
 - phon_no
 - availability_status
 - yers_of_work
 - langugs_known
-

Adventure_route

Stores trekking route information.

Fields:

- adv_rout_id
- rout_name
- destination
- difficulty_level
- capcity
- remening_spots
- rout_price
- journey_date
- track_status
- Status_open_close

- staf_fr_this_route

AdventureBooking

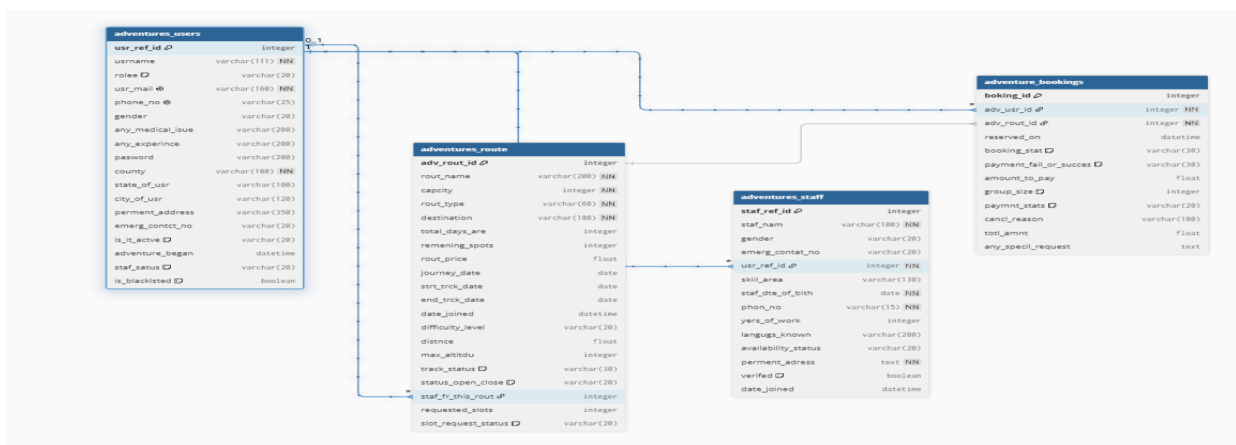
Stores user booking records.

Fields:

- boking_id
- adv_usr_id
- adv_route_id
- reserved_on
- booking_stat
- payment_fail_or_succes
- group_size
- totl_amnt

Relationships

- **One-to-Many:** AdventureUser → AdventureBooking
- **One-to-Many:** Adventure_route → AdventureBooking
- **One-to-Many:** AdventureUser (Staff) → Adventure_route
- **One-to-One:** AdventureUser → Adventure_Staff



6. Application Routes (instead of API Endpoints)

Since your project is a Flask web application and **does not expose REST APIs**, replace the API section with application routes.

Route	Method	Description
/login	GET, POST	User login
/register	GET, POST	User registration
/admin_dashboard	GET	Administrator dashboard
/staff_dashboard	GET	Staff dashboard
/user_dashboard	GET	User dashboard
/add_trek	GET, POST	Create a new trek
/manage_trek	GET	Manage trekking routes
/manage_users	GET	View and manage users
/manage_staff	GET	View and manage staff
/book_trek/<trek_id>	GET	Book a trekking route
/booking_table	GET	View all bookings
/user_booking_table	GET	View user's bookings
/logout	GET	Logout from the system